

Project planning phase-Template

Date	01 July 2026
Team ID	SWTID-2026-6128
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Emotion Detection & Learning Support Engine

1. Project Overview & Objectives

The primary objective of this project is to build an AI-driven engine that detects user emotions from text input and provides personalized learning support by utilizing machine learning, deep learning, and generative AI guidance.

2. Project Scope

- Emotion detection from textual input
- Multi-model prediction using **BiLSTM** and **BERT**
- AI-powered guidance using **Gemini API**
- Interactive dashboard using **Streamlit**
- Deployment for end-user interaction

3. Requirements Planning

Software Requirements

- Python 3.x
- Kaggle / Jupyter Notebook
- Streamlit
- TensorFlow / PyTorch
- Transformers Library
- Gemini API

Hardware Requirements

- Laptop / PC
- Minimum 8GB RAM

- GPU (preferred for training)
- Internet connection

4. Development Methodology

Incremental & Modular Agile Framework

- Divide project into smaller modules (Epics)
- Build and test each module separately
- Integrate step-by-step
- Continuous improvement after each phase

Project Modules / Epics

- **Epic 1:** Environment Setup & Dependency Configuration
- **Epic 2:** Kaggle Model Training & Integration
- **Epic 3:** Core Emotion Detection Pipeline Development
- **Epic 4:** AI-Powered Guidance & Regeneration Engine
- **Epic 5:** Streamlit UI Implementation
- **Epic 6:** User Interaction & Deployment Validation

5. Timeline & Expected Deliverables

Timeline Planning

Project Phase	Duration
Planning	1 Week
Data Preparation	1 Week
Model Training	2 Weeks
Pipeline Development	1 Week
UI Development	1 Week
Testing & Deployment	1 Week

Expected Deliverables

- Trained AI models
- Emotion classification system
- Guidance generation engine
- Streamlit web application

- Final deployed product

6. Risk Management & Success Criteria

Risk Mitigation Matrix

Identified Risks	Mitigation Strategies
Model accuracy issues	Regular testing
API failures	Backup models
Deployment errors	Error handling
Slow response time	Performance optimization

Key Success Criteria

- High emotion detection accuracy
- Fast response generation
- Smooth UI experience
- Successful deployment

This clean project planning phase template can be directly imported into your project documentation or converted into an infographic style matching your previous slides.